ALEXANDER LAROCHE

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Research Interests

- **Binary Evolution:** binary-stripped helium stars, mass transfer, post-interaction binaries, binary parameter inference, compact object companions, magnetic stars, Magellanic Clouds;
- **Astrostatistics:** generative models, variational auto-encoders, Gaia XP spectra, stellar abundances, carbon-enhanced metal-poor stars, OB stars, multimodal learning with time-series data, unresolved binaries;
- **Observations:** UV, optical and infrared photometry and spectroscopy, radial velocities;

Professional Appointments

Graduate Researcher, University of Toronto 2022–Present

Undergraduate Research Assistant, University of Toronto 2021–2022

Undergraduate Research Assistant, McGill University 2020–2021

Education

University of Toronto 2022–Present

- Ph.D. Astronomy & Astrophysics

McGill University 2018–2022

- B.Sc. Honours Physics, First Class Honours

Awards & Honors

Walter C. Sumner Memorial Fellowship, Walter C. Sumner Foundation 2026

$8,000 CAD; to up to 40 top doctoral students nationally in chemistry, physics, electrical engineering or computer science

Canada Graduate Scholarship – Doctoral (CGS-D), NSERC 2024

$120,000 CAD; to high-calibre doctoral students in the natural sciences and engineering

DSI Doctoral Student Fellowship, Data Sciences Institute, University of Toronto 2024

$75,000 CAD; to 14 doctoral students for multidisciplinary research in data science, co-supervised across disciplines

Canada Graduate Scholarship – Master's (CGS-M), NSERC 2023

$17,500 CAD; to high-calibre master's students in the natural sciences and engineering

Best Poster Award, 3,2,1: Massive Triples, Binaries and Mergers, KU Leuven 2023

for "An envelope lost in transport"

Graduate Entrance Scholarship, University of Toronto 2022

$5,000 CAD; to incoming graduate students in the David A. Dunlap Department of Astronomy & Astrophysics

Dean's Honours List, McGill University 2021

top 10% of the Faculty of Science

Undergraduate Student Research Award (USRA), NSERC 2021

$9,600 CAD; to undergraduate students to gain research experience in the natural sciences and engineering

Canada 150 Research Chair Undergraduate Research Grant, McGill University 2020

$6,000 CAD; for undergraduate research with Jonathan Sievers, Canada 150 Research Chair at McGill University

Charles River Laboratories Scholarship, McGill University 2020

$2,000 CAD; to two outstanding undergraduate students in the Faculty of Science, on the basis of academic merit

Wing Hing Chan Scholarship in Science, McGill University 2020

$500 CAD; to outstanding undergraduate students in the Faculty of Science, on the basis of academic merit

Publication Summary

8 papers, 3 as first author, 1 as second or third author; 103 citations; h-index $= 4$ (ADS Library)

Research Mentorship

Alejandro Ortega-Cruz Pietro (UofT UG, multimodal learning for variable stars) 2024–2025

Observatory Allocations

Principal Investigator:

ESO Very Large Telescope, X-SHOOTER, 6 hours 2026
Gemini South, GHOST, 5.8 hours 2026

Co-Investigator:

Hubble Space Telescope, COS, 7 orbits 2026
Gemini South, GMOS, 48.5 hours 2025–2026
Lick Observatory, Kast, 160 hours 2025–2026
Gemini South, GHOST, 15.5 hours 2024–2025
Keck Observatory, LRIS, 1.5 nights 2025
Magellan Telescopes, MagE, 6.5 nights 2023–2025

Teaching

Teaching Assistant, Introduction to Astrophysical Research, University of Toronto Fall 2026
Teaching Assistant, Introduction to Astrophysics, University of Toronto Winter 2026
Teaching Assistant, AST221: Stars and Planets, University of Toronto Fall 2025
Teaching Assistant, AST325/326: Introduction to/Practical Astronomy, University of Toronto Fall 2024
Teaching Assistant, AST222: Galaxies & Cosmology, University of Toronto Winter 2024
Teaching Assistant, AST199: Astronomy at the Frontier, University of Toronto Fall 2023
Teaching Assistant, AST201: Stars & Galaxies, University of Toronto Winter 2023, 2025
Teaching Assistant, AST101: The Sun and Its Neighbours, University of Toronto Fall 2022
Math and Physics Tutor, PAPER Tutor, Montreal 2019–2020

Professional Service

Journal Referee (MNRAS, ApJ) 2022–Present
Treasurer, UofT Graduate Astronomy Student Association 2023–Present
Astronomy Representative, UofT Graduate Student Union 2022–2023

Outreach

Filmographer, UofT AstroTours 2022–2023
Graduate School Panelist, UofT Astronomy Union 2022

Talks

Invited Talks and Seminars

Asteroseismology Group Meeting, KU Leuven July, 2025

TASTY Talk, University of Toronto (video)	March, 2025
Tea Talk, Carnegie Observatories	December, 2024
Tea Talk, California Institute of Technology	December, 2024
Institute Seminar, Institute of Science and Technology Austria	July, 2024

Contributed Talks

From Discovery to a Population: Benchmarking Stripped Stars and Companions, Leiden	July, 2024
New Computational Methods in Milky Way Dynamics and Structure, Bavaria	July, 2024
Rare Gems in Big Data, Tucson	May, 2024
Stable Mass Transfer in Binaries: From Onset to Remnants, New York	March, 2024
Canadian Astro-Particle Physics Summer Student Talk Competition, Sudbury	August, 2022
Stellar Stats Workshop, University of Toronto	May, 2022
Undergraduate Research Project Presentation, McGill University	December, 2021
NSERC USRA Poster Seminar, University of Toronto	August, 2021
Undergraduate Thesis Presentation, McGill University	April, 2021

Publication List (ADS Library)

First-Author Publications

3. **A. Laroche**, M. R. Drout, Y. Götberg, et al. (2026)
"Revealing Optically Hidden Companions to Intermediate-Mass Stripped Stars with Radial Velocities"
Submitted to ApJ

2. **A. Laroche**, J. S. Speagle (2025)
"Closing the Stellar Labels Gap: Stellar Label independent Evidence for [α/M] Information in Gaia BP/RP Spectra"
The Astrophysical Journal, 979, 5

1. **A. Laroche**, D. Gilman, X. Li, et al. (2022)
"Quantum fluctuations masquerade as haloes: bounds on ultra-light dark matter from quadruply imaged quasars"
Monthly Notices of the Royal Astronomical Society, 517, 1867

Second- and Third-Author Publications

1. T. Géron, **A. Laroche**, J. S. Speagle, et al. (2026)
"Unresolved Binary Systems in the Rubin Era I: An Autoencoder Framework for Binary Identification Applied to 47 Tucanae"
arXiv preprint

Contributing-Author Publications

4. L. A. C. van Son, N. Yamaguchi, P. Nagarajan, et al. (2026)
"Ongoing and Post-mass-transfer Binaries: A Living Catalog and Unified Review of Binary Mass-transfer Products"
The Astrophysical Journal Supplement Series, 286, 13

3. S. M. Grondin, M. R. Drout, J. Nordhaus, et al. (2026)
"A Framework for Linking Pre- and Post-Common Envelope Binary Properties with Star Clusters: The First Demonstration with a Massive White Dwarf+M Dwarf Binary in Alessi 12"
arXiv preprint

2. B. Ludwig, M. R. Drout, Y. Götberg, et al. (2026)
"The Stripped-star Ultraviolet Magellanic Cloud Survey (SUMS): The Ultraviolet Photometric Catalog and Stripped-star Candidate Selection"
The Astrophysical Journal, 999, 73

1. F. A. Evans, S. M. Grondin, C. S. Ye, et al. (2025)
"The ejection and detectability of high- and hyper-velocity stars by compact object binaries in globular clusters"
arXiv preprint

Other Publications

1. **A. Laroche**, J. S. Speagle (2023)
"Closing the stellar labels gap: An unsupervised, generative model for Gaia BP/RP spectra"
ICML 2023 Workshop on Machine Learning for Astrophysics